

Assignment 12: Fashion-MNIST Image Classification Using Neural Networks

Introduction

This project explores the use of a Feedforward Neural Network (FNN) for image classification using the Fashion-MNIST dataset. Fashion-MNIST contains 70,000 grayscale images of clothing items divided into 10 categories, including T-shirts, trousers, dresses, coats, shoes, and bags. The objective was to train a neural network capable of automatically classifying clothing images and evaluate its performance using multiple machine learning metrics. The neural network used in this project consists of an input layer, two hidden layers with ReLU activation functions, and a Softmax output layer for multi-class classification.

Methodology

Dataset Preparation

The Fashion-MNIST dataset was loaded using TensorFlow/Keras. The dataset contains 60,000 training images and 10,000 testing images. Sample images were visualized to understand the different classes and variations within the dataset.

Image Preprocessing

The image pixel values originally ranged from 0 to 255. To improve training performance, all pixel values were normalized by dividing them by 255, resulting in values between 0 and 1. The images were then flattened from 28×28 pixels into one-dimensional vectors of 784 features to serve as input to the feedforward neural network.

Model Architecture

The initial neural network architecture consisted of:

- Input Layer: 784 neurons
- Hidden Layer 1: 128 neurons with ReLU activation
- Hidden Layer 2: 64 neurons with ReLU activation
- Output Layer: 10 neurons with Softmax activation

The model was compiled using:

- Optimizer: Adam

- Loss Function: Sparse Categorical Crossentropy
- Evaluation Metric: Accuracy

The model was trained for 15 epochs using a batch size of 64 with a validation split of 20%.

Evaluation Metrics

The model was evaluated using:

- Accuracy
- Precision
- Recall
- F1-Score
- Classification Report
- Confusion Matrix

These metrics provide insight into both overall performance and individual class performance.

Results Analysis

Accuracy Analysis

The training accuracy increased steadily from approximately 81% to over 92%, demonstrating that the neural network successfully learned meaningful patterns from the training data. Validation accuracy increased from approximately 85% to 89% before stabilizing. The relatively small gap between training and validation accuracy indicates only minor overfitting and good generalization performance.

Loss Analysis

Training loss decreased consistently from approximately 0.54 to 0.20 over the 15 training epochs. Validation loss initially decreased before stabilizing around 0.31–0.34. Although minor fluctuations appeared in later epochs, the overall trend indicates effective learning with only mild overfitting.

Classification Report Analysis

The model achieved an overall test accuracy of 88%.

Best-performing classes:

- Trouser
- Sandal
- Bag
- Ankle Boot

These classes achieved very high precision, recall, and F1-scores because they possess distinctive visual characteristics.

Lower-performing classes:

- Shirt
- Pullover
- Coat

These classes were more frequently confused because they share similar shapes and textures. The model achieved macro-average and weighted-average F1-scores of approximately 0.88, indicating strong and balanced performance across all categories.

Confusion Matrix Analysis

The confusion matrix revealed that most predictions were concentrated along the diagonal, indicating correct classification. The highest classification accuracy was observed for Trouser, Sandal, Sneaker, Bag, and Ankle Boot classes. The most common classification errors occurred between Shirts, Pullovers, Coats, and T-shirts due to their visual similarity. Despite these challenges, the overall classification performance was strong and consistent.

Model Improvements

To improve performance, the neural network architecture was expanded:

- Hidden Layer 1: 128 → 256 neurons
- Hidden Layer 2: 64 → 128 neurons

The improved model increased the total number of trainable parameters to 235,146, allowing the network to learn more complex image features. Data augmentation techniques such as rotation, zooming, and shifting were also explored to improve model robustness and reduce overfitting. These techniques help expose the model to a wider variety of image variations during training.

Practical Applications and Challenges

Practical Application

A practical use case for this model is in the fashion retail industry. An online retailer could automatically classify clothing items uploaded by vendors and place them into appropriate product categories. This would reduce manual effort and improve inventory management.

Deployment Challenges

Several challenges must be considered when deploying the model:

- Scalability when processing large product catalogs.
- Real-time prediction requirements for e-commerce systems.
- Integration with existing inventory and database systems.
- Continuous retraining as new products and fashion trends emerge.
- Maintaining classification accuracy on previously unseen clothing styles.

Conclusion

This project successfully demonstrated the application of a feedforward neural network to the Fashion-MNIST image classification task. The model achieved approximately 88% test accuracy and showed strong performance across most clothing categories.

Normalization significantly improved training efficiency, while experimentation with larger hidden layers demonstrated potential performance gains. Future improvements could include implementing Convolutional Neural Networks (CNNs), additional data augmentation techniques, and hyperparameter optimization to further increase

classification accuracy and reduce misclassification among visually similar clothing categories.

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